

Spinning Pyramid

one idea behind all this logic spinning

you have a dot in 3D space, you do math on it (mostly or all dot product of matrix of their rotation matrixes $\text{rotX}, \text{rotY}, \text{rotZ}$), it appears on 2D screen, do that for every dot on the shape, 60 times a second (could be more I am doing 60 times change by usleep()), your brain sees a spinning pyramid, that's literally the whole program. everything else is just details of that one idea

the cube face vs triangle face problem

a cube face is a square, two loops that work for a square because both x and y can freely go to their maximum but if we try this in a triangle, if you let x and y both go to their max it will draw a square not a triangle

barycentric the mvp

The concept is i.e: you are a mix of the 3 corners

$$\text{point} = u \cdot A + t \cdot B + s \cdot C$$

$$u + t + s \text{ must always} = 1$$

It's like tug of war between 3 people standing at the corners, wherever you land is where the three tugs balance out, the loop is just one rule:

s cannot go past what t did not use

Pseudo logic is this

t from 0 to 1

s from 0 to $(1 - t)$

$$u = 1 - t - s$$

if t already used 0.6, s can only go to 0.4 max that slanted edge of the triangle

The end conclusion: the pyramid never moves, we take each dot, do math, ask where does it land on screen now, draw a char there, doing that 60 times a second and your eye sees a spinning pyramid